

Covered: Vitali's Construction, Lebesgue measure over linear transformations, Regularity Property

1 Vitali's Construction

Aim: To demonstrate, using the Axiom of Choice, the existence of subsets of $\mathbb{R}$ which are not in $\mathcal{B}_{\mathbb{R}}$.

We proceed on the unit interval $((0, 1], \mathcal{B}_{(0,1]}, \lambda)$. Define the operation

$$\oplus : (0, 1] \times (0, 1] \longrightarrow (0, 1] \quad \text{with,} \quad x \oplus y = \begin{cases} x + y, & x + y \leq 1, \\ x + y - 1, & \text{otherwise.} \end{cases}$$

Then $((0, 1], \oplus)$ is an abelian group with 1 as the additive identity. Also, $\mathbb{Q} \cap (0, 1]$ is a subgroup of $((0, 1], \oplus)$.

Define a relation $\sim$ on $(0, 1]$ by

$$x \sim y \iff \exists q \in \mathbb{Q} \cap (0, 1] \text{ such that } y = x \oplus q.$$

Theorem 1.1. $\sim$ is an equivalence relation.

Proof. *Reflexive:* $x = x \oplus 1 \implies x \sim x$

Symmetric: Suppose $x \sim y$. Then $y = x \oplus q$ for some $q \in \mathbb{Q} \cap (0, 1]$. Since $\mathbb{Q} \cap (0, 1]$ is a group, there exists $q' \in \mathbb{Q} \cap (0, 1]$ such that $q \oplus q' = 1$. Hence $x = y \oplus q'$ and therefore $y \sim x$.

Transitive: Suppose $x \sim y$ and $y \sim z$. Then $x = y \oplus q$ with $y = z \oplus q'$ for some $q, q' \in \mathbb{Q} \cap (0, 1]$. Thus

$$x = (z \oplus q') \oplus q = z \oplus (q' \oplus q),$$

and since $q' \oplus q \in \mathbb{Q} \cap (0, 1]$, we have $x \sim z$. □

Let $\{A_\alpha\}_{\alpha \in I}$ be the collection of equivalence classes of $\sim$. Thus, if $x \sim y$, then there exists a unique $\alpha \in I$ such that $x, y \in A_\alpha$.

Nothing can be said about the sizes of the sets A_α a priori, but $\{A_\alpha\}_{\alpha \in I}$ is a nonempty collection of nonempty, pairwise disjoint subsets of $(0, 1]$.

By the Axiom of Choice, $\exists E = \{x_\alpha \mid \alpha \in I\}$ such that

$$x_\alpha \in A_\alpha \quad \forall \alpha \in I.$$

Thus E contains exactly one representative from each equivalence class.

Clearly $E \subseteq (0, 1]$. For $q \in \mathbb{Q} \cap (0, 1]$, define

$$q \oplus E := \{q \oplus x \mid x \in E\}.$$

Theorem 1.2.

$$\bigcup_{q \in \mathbb{Q} \cap (0, 1]} (q \oplus E) = (0, 1],$$

and the sets $\{q \oplus E\}_{q \in \mathbb{Q} \cap (0, 1]}$ are pairwise disjoint.

Proof. Suppose

$$q \oplus E \cap q' \oplus E \neq \emptyset$$

for some $q, q' \in \mathbb{Q} \cap (0, 1]$. Then there exist $\alpha, \alpha' \in I$ such that $q \oplus x_\alpha = q' \oplus x_{\alpha'}$.

Hence

$$x_\alpha = (q^{-1} \oplus q') \oplus x_{\alpha'},$$

where $q^{-1} \oplus q' \in \mathbb{Q} \cap (0, 1]$. Therefore $x_\alpha \sim x_{\alpha'}$.

Since x_α and $x_{\alpha'}$ are representatives of their equivalence classes,

$$\alpha = \alpha' \implies x_\alpha = x_{\alpha'}.$$

Cancelling x_α gives

$$q = q'.$$

Thus

$$q \oplus E \cap q' \oplus E \neq \emptyset \implies q = q',$$

so the sets $q \oplus E$ are pairwise disjoint.

$$x \in (0, 1] = \bigcup_{\alpha \in I} A_\alpha \implies \exists! \alpha \in I \text{ such that } x \in A_\alpha.$$

Hence, for some $q \in \mathbb{Q} \cap (0, 1]$, with $x = q \oplus x_\alpha$ and therefore $x \in q \oplus E$. Thus

$$(0, 1] = \bigcup_{q \in \mathbb{Q} \cap (0, 1]} (q \oplus E).$$

□

Suppose $E \in \mathcal{B}_{(0,1]}$. Then

$$q \oplus E \in \mathcal{B}_{(0,1]} \quad \forall q \in \mathbb{Q} \cap (0, 1],$$

and, by translation invariance of Lebesgue measure,

$$\lambda(q \oplus E) = \lambda(E).$$

Since the sets $\{q \oplus E\}_{q \in \mathbb{Q} \cap (0, 1]}$ are pairwise disjoint,

$$\begin{aligned} 1 = \lambda((0, 1]) &= \lambda\left(\bigcup_{q \in \mathbb{Q} \cap (0, 1]} (q \oplus E)\right) \\ &= \sum_{q \in \mathbb{Q} \cap (0, 1]} \lambda(q \oplus E) = \sum_{q \in \mathbb{Q} \cap (0, 1]} \lambda(E), \end{aligned}$$

Since $\lambda(E)$ can either be 0 or finite, the above sum can be either 0 or infinite. Contradiction. Hence

$$\boxed{E \notin \mathcal{B}_{(0,1]}}.$$

Let $(\mathbb{R}, \mathcal{B}_{\mathbb{R}})$ be the Borel measurable space, where $\mathcal{B}_{\mathbb{R}}$ is countably generated, and let $\Omega \neq \emptyset$.

A measurable map $T : \Omega \rightarrow \mathbb{R}$ is called a *statistic*. The σ -algebra generated by T is

$$\sigma(T) := \{T^{-1}(B) : B \in \mathcal{B}_{\mathbb{R}}\}.$$

Since $\mathcal{B}_{\mathbb{R}}$ is countably generated, $\sigma(T)$ is also countably generated.

Exercise 1.3. If $\mathcal{F}$ is a countably generated σ -algebra on Ω , then there exists a measurable map $T : \Omega \rightarrow \mathbb{R}$ such that $\sigma(T) = \mathcal{F}$.

Theorem 1.4. Fix $d \geq 1$. Let $T : \mathbb{R}^d \rightarrow \mathbb{R}^d$ be a nonsingular linear transformation. Then, for every $B \in \mathcal{B}_{\mathbb{R}^d}$, $T(B) \in \mathcal{B}_{\mathbb{R}^d}$ and

$$\lambda_d(T(B)) = |\det T| \lambda_d(B)$$

Proof. Since T is nonsingular, $T(B) = (T^{-1})^{-1}(B)$. Since T^{-1} is continuous, T^{-1} is measurable, and therefore $T(B) \in \mathcal{B}_{\mathbb{R}^d}$.

It is enough to verify the formula for a collection of elementary linear transformations which generate every nonsingular linear transformation. Indeed,

$$T = T_1 \circ T_2 \circ \cdots \circ T_N$$

for some N , where the T_i 's are of the following forms:

1. Scalar multiplication:

$$\begin{pmatrix} x_1 \\ \vdots \\ x_d \end{pmatrix} \mapsto \begin{pmatrix} \alpha x_1 \\ \vdots \\ \alpha x_d \end{pmatrix}, \quad \alpha \in \mathbb{R} \setminus \{0\}.$$

2. Permutation of coordinates:

$$\begin{pmatrix} x_1 \\ \vdots \\ x_d \end{pmatrix} \mapsto \begin{pmatrix} x_{\pi(1)} \\ \vdots \\ x_{\pi(d)} \end{pmatrix}, \quad \pi \in S_d.$$

3. Elementary shear:

$$\begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_d \end{pmatrix} \mapsto \begin{pmatrix} x_1 + x_2 \\ x_2 \\ \vdots \\ x_d \end{pmatrix}.$$

For the scalar map,

$$\lambda_d(\alpha B) = |\alpha|^d \lambda_d(B).$$

For permutations and elementary shears, Lebesgue measure is preserved. The general result then follows by composition. $\square$

Theorem 1.5. If the formula

$$\lambda_d(T(B)) = |\det T| \lambda_d(B)$$

holds for T and T' , then it also holds for $T \circ T'$.

Proof.

$$\begin{aligned} \lambda_d((T \circ T')(B)) &= \lambda_d(T(T'(B))) \\ &= |\det T| \lambda_d(T'(B)) \\ &= |\det T| |\det T'| \lambda_d(B) = |\det(T \circ T')| \lambda_d(B). \end{aligned}$$

$\square$

Remark 1.6. For $\alpha > 0$ and $B \in \mathcal{B}_{\mathbb{R}^d}$, $\lambda_d(\alpha B) = \alpha^d \lambda_d(B)$.

Theorem 1.7 (Markov's inequality). Let $(\Omega, \mathcal{F}, \mu)$ be a measure space and let $f : \Omega \rightarrow [0, \infty]$ be measurable. Then, for every $\epsilon > 0$,

$$\mu(\{\omega \in \Omega : f(\omega) \geq \epsilon\}) \leq \frac{1}{\epsilon} \int_{\Omega} f \, d\mu.$$

Theorem 1.8 (Regularity property of Lebesgue measure). Let $B \in \mathcal{B}_{\mathbb{R}}$ with $\lambda(B) < \infty$. Then, for every ϵ , there exist a compact set K_ϵ and an open set U_ϵ such that

$$K_\epsilon \subseteq B \subseteq U_\epsilon$$

and

$$\lambda(U_\epsilon \setminus K_\epsilon) < \epsilon.$$

Corollary 1.9. For every $B \in \mathcal{B}_{\mathbb{R}}$ and every $\epsilon > 0$, there exist a closed set C_ϵ and an open set U_ϵ such that

$$C_\epsilon \subseteq B \subseteq U_\epsilon$$

and

$$\lambda(U_\epsilon \setminus C_\epsilon) < \epsilon.$$

proof pending.